

Thompson - CS 505 Module Seven Activity Boolean Algebra

Complete this template by replacing the bracketed text with the relevant information.

1. Use Boolean algebra to simplify the Boolean expressions provided:

A. $F(A, B, C) = (A'BC')' + A(B' + C)$

Answer:

$$(A'BC')' = (A')' + B' + (C')'$$

$$= A + B' + C$$

$$F(A, B, C) = A + B' + C + A(B' + C):$$

Using the Absorption Law: $A + AB = A$ means:

$$A + B' + C + A(B' + C) = A + B' + C$$

$$\textbf{Simplified Expression} = A + B' + C$$

B. $A'B + AB'$

Answer:

This is as simplified as it gets. It could be rewritten as $(A + B)(A' + B')$ or simply *XOR* since this is an exclusive OR expression.

C. $(A + B)(A' + C)$

Answer:

$$AA' + AC + BA' + BC$$

$$0 + AC + A'B + BC$$

$$AC + A'B \text{ (Removed } BC \text{ with Consensus Theorem)}$$

$$\textbf{Simplified Expression} = (A + B)(A' + C) = AC + A'B$$

D. $(A + B)(A + C)(B + C)$

Answer:

$$= (AA + AC + AB + BC)(B + C)$$

$$= (A + BC)(B + C): AC \text{ and } AB \text{ were removed with absorption.}$$

$$= AB + BBC + AC + BCC$$

$$= AB + BC + AC$$

2. Use Boolean algebra to prove the Boolean equivalence provided:

$$(A + B)(A' + C) = AC + A'B$$

Answer:

$$AA' + AC + A'B + BC = AC + A'B$$

$$0 + AC + A'B + BC = AC + A'B$$

$$\text{By the Consensus Theorem: } XY + X'Z + YZ = XY + X'Z$$

$$\text{Here: } X = A, Y = C, Z = B$$

$$\text{So: } AC + A'B + BC = AC + A'B$$

$$\text{Therefore } (A + B)(A' + C) = AC + A'B$$

3. Implement the Boolean function provided, using only NAND gates.

$$F(A, B, C) = AB' + A'C$$

$$= \overline{\overline{AB' + A'C}}$$

$$= \overline{(AB') \overline{(A'C)}}$$

$$= (A \text{ NAND } (B \text{ NAND } B)) \text{ NAND } ((A \text{ NAND } A) \text{ NAND } C)$$

4. Use Boolean algebra laws to simplify the truth table provided:

A B C F

0 0 0 1

0 0 1 0

0 1 0 0

0 1 1 1

1 0 0 0

1 0 1 1

1 1 0 1

1 1 1 1

Answer:

$$F(A, B, C) = A'B'C' + A'BC + AC + AB$$

5. Draw a logic circuit diagram for the simplified expression of the truth table provided:

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Answer

$A \xrightarrow{\text{[NOT]}} A'$
 $B \xrightarrow{\text{[NOT]}} B'$
 $C \xrightarrow{\text{[NOT]}} C'$

$A' \xrightarrow{\text{[AND]}} T1$
 $B' \xrightarrow{\text{[AND]}} T2$
 $C' \xrightarrow{\text{[AND]}} T3$

$A \xrightarrow{\text{[AND]}} T4$
 $B \xrightarrow{\text{[AND]}} T5$

$A \xrightarrow{\text{[AND]}} T6$
 $C \xrightarrow{\text{[AND]}} T7$

$$T1 = A'B'C'$$

$$T2 = A'BC$$

$$T3 = AB$$

$$T4 = AC$$

$T1 \xrightarrow{\text{[OR]}}$

$T2 \xrightarrow{\text{[OR]}} F$

$T3 \xrightarrow{\text{[OR]}}$

$T4 \xrightarrow{\text{[OR]}}$

6. Create a truth table for the Boolean expressions provided:

A. $F(A, B, C) = (A \text{ OR } B') \text{ AND } (A' \text{ OR } C)$

Answer:

row	A	B	C	A'	B'	A + B'	A' + C	(A + B')(A' + C)	F(A, B, C)
0	0	0	0	1	1	1	1	1	1
1	0	0	1	1	1	1	1	1	1
2	0	1	0	1	0	0	1	0	0
3	0	1	1	1	0	0	1	0	0
4	1	0	0	0	1	1	0	0	0
5	1	0	1	0	1	1	1	1	1
6	1	1	0	0	0	1	0	0	0
7	1	1	1	0	0	1	1	1	1

$$F(A, B, C) = \Sigma(0, 1, 5, 7)$$

B. $(A, B, C) = \Sigma(0, 1, 2, 4, 7)$

Answer:

row	A	B	C	F
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

7. Create a truth table to show the output for all possible input combinations for the Boolean expressions provided:

A. $F(A, B) = A \text{ AND } B$

Answer:

row	A	B	F
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$$F(A, B) = \Sigma(3)$$

B. $F(A, B) = A \text{ OR } B$

Answer:

row	A	B	F
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$$F(A, B) = \Sigma(1, 2, 3)$$

C. $F(A, B) = \text{NOT } A$

Answer:

row	A	B	F
0	0	0	1
1	0	1	1
2	1	0	0
3	1	1	0

$$F(A, B) = \Sigma(0, 1)$$

D. $F(A, B) = A'B + AB'$

Answer:

row	A	B	A'	B'	A'B	AB'	F
0	0	0	1	1	0	0	0
1	0	1	1	0	1	0	1
2	1	0	0	1	0	1	1
3	1	1	0	0	0	0	0

$$F(A, B) = \Sigma(1, 2)$$

E. $F(A, B, C) = A'BC + AB'C' + ABC$

Answer:

row	A	B	C	A'	B'	C'	A'BC	AB'C'	ABC	F
0	0	0	0	1	1	1	0	0	0	0
1	0	0	1	1	1	0	0	0	0	0
2	0	1	0	1	0	1	0	0	0	0
3	0	1	1	1	0	0	1	0	0	1
4	1	0	0	0	1	1	0	1	0	1
5	1	0	1	0	1	0	0	0	0	0
6	1	1	0	0	0	1	0	0	0	0
7	1	1	1	0	0	0	0	0	1	1

$$F(A, B, C) = \Sigma(3, 4, 7)$$